

Bernard Lee

DATA TECHNOLOGY & AI LEADER

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Technology leader spanning enterprise data architecture, AI, analytics, and digital delivery. Combines hands-on engineering with architecture governance, business-case development, and cross-functional leadership—translating complex business needs into secure, scalable data platforms with measurable operational and commercial outcomes.

RECENT DATA LEADERSHIP

Hong Kong Monetary Authority Technology Innovation Lead

Jul 2025–Present

- Re-architected and migrated a four-service Azure research data-fetching solution into one containerized AWS web application within a month, supporting four sources while preserving performance and improving usability and error visibility.
- Built an end-to-end natural-language analytics PoC in three months for five pilot users, enabling permission-controlled warehouse querying and AI-generated visualizations.
- Orchestrated schema-discovery, RAG, SQL-generation, validation, execution, and visualization agents, targeting a shift from an estimated three-day Data CoE handoff to near-real-time self-service.
- Developed a multi-agent risk-profiling PoC that processed up to thousands of external articles per run and attached confidence scores and evidence to semantic risk tags.
- Led the Architecture Review Board-approved design of an enterprise API and AI gateway for up to 100M transactions, covering semantic caching, observability, routing, fallback, cost, and model controls.
- Coordinated vendor teams of up to 15–20 engineers while defining authentication, security, and scalable data-access requirements across Kubernetes and VM services.

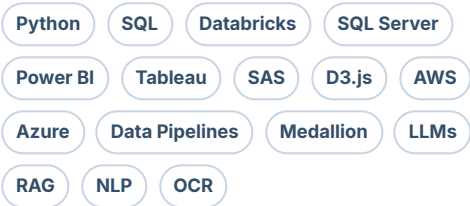
Hong Kong Monetary Authority Data Architect

Nov 2023–Jun 2025

- Reviewed data flows, architecture diagrams, and proposals for 10–15 projects, providing Architecture Review Board recommendations on secure, scalable data design.
- Defined the organization's medallion data architecture, approved as part of the enterprise reference architecture, with reusable patterns across ingestion, transformation, storage, and consumption.
- Established secure proxy-based web-scraping and monitoring patterns adopted by at least five teams and projects retrieving external data.
- Developed Python reference implementations and concurrent ingestion pipelines that reduced external-data retrieval from five hours to under one.
- Delivered two webinars and an in-person session on data architecture and AI for audiences ranging from 25 to more than 300 people.

DATA PROFILE

Core data stack



Architecture & governance

Enterprise data architecture · Architecture Review Board · Data governance · Medallion design · Data-access controls · Secure ingestion · API and AI gateways · Cloud migration · Reference implementations

Data & AI delivery

Data ingestion · Warehousing · BI · Claims analytics · Financial-crime analytics · Natural-language-to-SQL · Multi-agent orchestration · NLP · OCR · Executive reporting

Education & details

BEng Electrical & Electronics Engineering
Sheffield Hallam University
First City University College · 2010

Hong Kong Permanent Resident
English · Cantonese

Best-fit focus

Enterprise data architecture · Data platforms · Data governance · Data engineering · AI & analytics · Financial and insurance data

DATA & ANALYTICS EXPERIENCE

MCM Worldwide Jul 2022–Oct 2023
Principal Data Architect

- Delivered Global Finance and CEO dashboards covering sales indicators and KPIs across five countries, supported by pipelines spanning the enterprise SQL Server data warehouse.
- Created optimized warehouse views that improved data accuracy and reduced Power BI query response from minutes to seconds.
- Introduced modular design patterns across 20–30 dashboard and query-performance components, supporting a broader framework of up to 20 dashboards.
- Led two BI specialists and five workshops for 30–50 global participants on dashboard usability, data availability, and adoption.

AIA Hong Kong & Macau Jan 2019–Jul 2022
Associate Director · Senior Manager, Health & Wellness

- Delivered five customer-partnered digital health programs for three enterprise clients representing more than 6,000 employees.
- Combined claims, health-program, engagement, and participant data for weekly reporting and quarterly wellness-planning analysis.
- Increased program adoption by 25 percentage points; health-program analysis associated participation with 5–10% lower claims costs.
- Productionized a 90%-accurate Azure OCR capability for scoped claims, replacing 25% of manual claims QA workload, and developed NLP-based keyword categorization for health data.
- Built business cases covering US\$3M in analytics capability, US\$1M in application development, and multiple US\$60K PoCs.
- Coordinated internal data, IT, finance, and business teams plus 3–4 implementation partners across full user journeys and technical delivery.

KPMG Malaysia & KPMG US Dec 2016–Dec 2018
Advisory Manager · Consultant, US Secondment

- Led analytics delivery across six Malaysian engagements spanning banking investigations, AML reviews, and manufacturing analytics.
- Built a pure-Python NLP framework to ingest, parse, normalize, classify, search, and flag approximately 25M Bloomberg, Reuters, and email messages.
- Reused the framework to finish a later investigation in one month against a three-month allocation, reducing delivery time by approximately 67%.
- During a US secondment, analyzed approximately 5M transactions plus semi-structured SWIFT and CHIPS messages for two major banks.
- Built reusable Python assessment and SQL-generation modules for hardened environments and trained five colleagues in their use.

AIA Bhd. Nov 2012–Nov 2016
Data Analyst

- Led procurement and implementation of the first cross-functional Tableau deployment, initially supporting approximately 50 users across reporting, claims, and pricing.
- Built 10+ loss-ratio, executive, corporate-health, claims-monitoring, and operations reports for major clients and business stakeholders.
- Designed extraction, processing, and visualization pipelines across four source systems covering claims, client profiles, premiums, and transactions.
- Accelerated recurring report production by up to 10× through Python, SQL, and VBA automation.
- Used operational reporting and resource planning to reduce average call-centre waiting time by 30% within three months.

CAPABILITY SUMMARY

Enterprise architecture

Data strategy and reference architecture; project design review; hybrid-cloud patterns; data governance; security and access; reusable standards; vendor coordination.

Data engineering

Concurrent ingestion; web and external-data retrieval; ETL and transformation; warehouse views; multi-source pipelines; performance optimization; cloud migration.

Analytics & BI

Executive and finance dashboards; claims and wellness analytics; engagement reporting; financial investigations; AML analytics; operational planning; global BI adoption.

AI for data

Natural-language analytics; schema discovery; RAG; SQL generation and validation; multi-agent research; semantic tagging; NLP; OCR; confidence and evidence design.

Leadership scope

Architecture governance for 10–15 projects · Vendor teams of 15–20 engineers · Two BI direct reports · Workshops for 30–50 global participants · Knowledge sessions reaching 300+ attendees

Commercial scope

Business cases covering US\$3M in analytics capability, US\$1M in application development, and multiple US\$60K proof-of-concepts.